Machine Learning A (2026)
Home Assignment 2

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1 Preprocessing (35 points)

1.1 Importance of Preprocessing (8 points)

Using $K-NN$ i would define an input space with two variables $\mathcal{X} = \{(Age_in_years, Income_in_thousands_of_US)\}$ and an output space of weather the person defaults or not $\mathcal{Y} = \{yes, no\}$. From the table below we can derive the training data: $S = \{(47, 35, no), (22, 40, yes)\}$.

Person	Age in years	Income in thousands of USD	defaults
A	47	35	no
B	22	40	yes
C	21	36	

We want to predict weather person C defaults or not, to determine if person C should get a credit card. Using the euclidean norm, we can determine which person, person C is mostly identical to:

Distance to person A:

$$dist(A, C) = \sqrt{(47-21)^2 + (35000-36000)^2} \approx 1000.34.$$

Distance to person B:

$$dist(B, C) = \sqrt{(22-21)^2 + (40000-36000)^2} \approx 4000.00.$$

Thus, when measured in thousands of USD, person C looks more like person A (that did not default) than person B (that did default). According to this, the bank should therefore give person C a credit card.

Now let's try to calculate the euclidean norm again. This time assuming the income is given in USD:

Distance to person A:

$$dist(A, C) = \sqrt{(47-21)^2 + (35-36)^2} \approx 26.02.$$

Distance to person B:

$$dist(B, C) = \sqrt{(22-21)^2 + (40-36)^2} \approx 4.12.$$

This time person C looks more like person B than person A: The bank should not give person C a credit card. This is because the numerical difference between their incomes are significantly lower, when we measure ind USD instead of thousands of USD.

1.2 Input Centering (9 points)

1.2.1 (a)

Given the input matrix:

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \\ \vdots \\ \boldsymbol{x}_N^T \end{bmatrix},$$

and γ defined as:

$$\gamma = \boldsymbol{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^T,$$

i want to show that:

$$\boldsymbol{Z} = \gamma \boldsymbol{X},$$

where $\boldsymbol{Z}^T = \begin{bmatrix} \boldsymbol{z}_1 & \boldsymbol{z}_2 & \cdots & \boldsymbol{z}_N \end{bmatrix}$, where $\boldsymbol{z}_n = \boldsymbol{x}_n - \bar{\boldsymbol{x}}$.

Firstly i rewrite $\boldsymbol{Z}$, so it is expressed only in terms of $\boldsymbol{X}$ and $\mathbf{1}$:

$$\begin{aligned} \boldsymbol{Z} &= \begin{bmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \\ \vdots \\ \boldsymbol{x}_N^T \end{bmatrix} - \begin{bmatrix} \bar{\boldsymbol{x}}^T \\ \bar{\boldsymbol{x}}^T \\ \bar{\boldsymbol{x}}^T \\ \bar{\boldsymbol{x}}^T \end{bmatrix}, \\ &= \boldsymbol{X} - \mathbf{1}\bar{\boldsymbol{x}}^T, \\ &= \boldsymbol{X} - \mathbf{1}\frac{1}{N}\mathbf{1}^T\boldsymbol{X}, \\ &= \boldsymbol{X} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\boldsymbol{X}. \end{aligned}$$

This means, that i need to show:

$$\mathbf{Z} = \gamma \boldsymbol{X} = \boldsymbol{X} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\boldsymbol{X}.$$

Substituting in the expression for γ, i get:

$$\boldsymbol{Z} = \left(\boldsymbol{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\boldsymbol{X} = \boldsymbol{X} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\boldsymbol{X}.$$

The expression $\boldsymbol{X} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\boldsymbol{X}$ factorizes to $\left(\boldsymbol{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\right)\boldsymbol{X}$, meaning that the equality shown above is true. Thus, i have shown that:

$$\boldsymbol{Z} = \gamma \boldsymbol{X}.$$

1.2.2 (b)

To determine that rank($\boldsymbol{Z}$) $< N$, i use that the rank of the columns and the rank of the rows are equal. Consider the rank of the rows in the matrix $\boldsymbol{Z}$:

$$\boldsymbol{Z} = \begin{bmatrix} \boldsymbol{z}_1^T \\ \boldsymbol{z}_2^T \\ \vdots \\ \boldsymbol{z}_N^T \end{bmatrix}.$$

Since $\boldsymbol{z}_n$is constructed as the deviations of $\boldsymbol{x}_n$from the mean, $\boldsymbol{z}_n = \boldsymbol{x}_n - \bar{\boldsymbol{x}}$, the N rows of $\boldsymbol{Z}$ must be linearly dependent:

$$\sum_{n=1}^{N} \left(\boldsymbol{x}_n - \frac{1}{N} \sum_{n=1}^{N} \boldsymbol{x}_n \right) = \sum_{n=1}^{N} \boldsymbol{x}_n - N \frac{1}{N} \sum_{n=1}^{N} \boldsymbol{x}_n,$$

$$= \sum_{n=1}^{N} \boldsymbol{x}_n - \sum_{n=1}^{N} \boldsymbol{x}_n,$$

$$= 0.$$

Since at least the N rows of $\boldsymbol{Z}$ are linearly dependent, the rank of the rows in $\boldsymbol{Z}$ must be smaller than N (the rank is the maximum number of linearly independent rows).

1.3 Input Whitening (18 points)

1.3.1 (a)

The variance of x_1 is 1 since $\text{Var}(\hat{x}_1) = 1$, and $x_1 = \hat{x}_1$. The variance of x_2 is given by:

$$\text{Var}(x_2) = \text{Var}\left(\sqrt{1-\epsilon^2}\hat{x}_1 + \epsilon \hat{x}_2 \right),$$

$$= \text{Var}\left(\sqrt{1-\epsilon^2}\hat{x}_1 \right) + \text{Var}(\epsilon \hat{x}_2) + 2\text{Cov}\left(\sqrt{1-\epsilon^2}\hat{x}_1, \epsilon \hat{x}_2 \right).$$

Since $\hat{x}_1$ and $\hat{x}_2$ are independent, the covariance is 0, $\text{Cov}\left(\sqrt{1-\epsilon^2}\hat{x}_1, \epsilon \hat{x}_2 \right) = 0$. Using this gives me:

$$\text{Var}(x_2) = \text{Var}\left(\sqrt{1-\epsilon^2}\hat{x}_1 \right) + \text{Var}(\epsilon \hat{x}_2).$$

Since $\text{Var}(bX) = b^2\text{Var}(X)$, we can rewrite the expression:

$$\text{Var}(x_2) = \left(\sqrt{1-\epsilon^2} \right)^2 \text{Var}(\hat{x}_1) + \epsilon^2 \text{Var}(\hat{x}_2),$$

$$= (1-\epsilon^2)\text{Var}(\hat{x}_1) + \epsilon^2 \text{Var}(\hat{x}_2).$$

I know that $\text{Var}(\hat{x}_1) = \text{Var}(\hat{x}_2) = 1$. Using this yields the variance of x_2:

$$\text{Var}(x_2) = 1 - \epsilon^2 + \epsilon^2,$$

$$\text{Var}(x_2) = 1.$$

To calculate the covariance of x_1 and x_2, i use the following formula:

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y).$$

Using the formula yields:

$$\text{Cov}(x_1, x_2) = E\left(\hat{x}_1 \cdot \left(\sqrt{1-\epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2\right)\right) - E(\hat{x}_1)E\left(\sqrt{1-\epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2\right).$$

Since both $\hat{x}_1$ and $\hat{x}_2$ have zero mean, we get:

$$\text{Cov}(x_1, x_2) = E\left(\hat{x}_1 \cdot \left(\sqrt{1-\epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2\right)\right),$$

$$= E\left(\hat{x}_1^2\sqrt{1-\epsilon^2} + \hat{x}_2\hat{x}_1\epsilon\right),$$

$$= \sqrt{1-\epsilon^2}E(\hat{x}_1^2) + \epsilon E(\hat{x}_1\hat{x}_2).$$

Using that $E(\hat{x}_1) = 0$,$E(\hat{x}_1^2)$ can be rewritten to $E(\hat{x}_1^2) = E(\hat{x}_1^2) - E(\hat{x}_1)^2 = \text{Var}(\hat{x}_1) = 1$, which yields:

$$\text{Cov}(x_1, x_2) = \sqrt{1-\epsilon^2} + \epsilon E(\hat{x}_2\hat{x}_1).$$

Since Since $\hat{x}_1$ and $\hat{x}_2$ are independent both with zero mean, $E(\hat{x}_1\hat{x}_2)$ can be rewritten to $E(\hat{x}_1\hat{x}_2) = E(\hat{x}_1)E(\hat{x}_2) = 0$:

$$\text{Cov}(x_1, x_2) = \sqrt{1-\epsilon^2}.$$

1.3.2 (b)

To obtain (w_1, w_2) as functions of $(\hat{w}_1, \hat{w}_2)$ i start by plugging the expression for $\hat{x}_1$ into the expressions for x_2, and rewrite $\hat{x}_2$ as a function of (x_1, x_2):

$$x_2 = \sqrt{1-\epsilon^2}x_1 + \epsilon\hat{x}_2,$$

$$\Leftrightarrow \epsilon\hat{x}_2 = x_2 - \sqrt{1-\epsilon^2}x_1,$$

$$\Leftrightarrow \hat{x}_2 = \frac{x_2 - \sqrt{1-\epsilon^2}x_1}{\epsilon}.$$

Next i plug the above expression for $\hat{x}_2$ and the expression for $\hat{x}_1$ into the function $f(\hat{\boldsymbol{x}})$:

$$f(\hat{\boldsymbol{x}}) = \hat{w}_1\hat{x}_1 + \hat{w}_2\hat{x}_2,$$

$$\Leftrightarrow f(\boldsymbol{x}) = \hat{w}_1 x_1 + \hat{w}_2\frac{x_2 - \sqrt{1-\epsilon^2}x_1}{\epsilon},$$

$$= \left(\hat{w}_1 - \hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right) x_1 + \hat{w}_2 \frac{x_2}{\epsilon},$$
$$= \left(\hat{w}_1 - \hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right) x_1 + \frac{\hat{w}_2}{\epsilon} x_2.$$

From the expression above we can infer $w_1(\hat{w}_1, \hat{w}_2)$ and $w_2(\hat{w}_2)$:

$$w_1(\hat{w}_1, \hat{w}_2) = \hat{w}_1 - \hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon},$$

$$w_2(\hat{w}_2) = \frac{\hat{w}_2}{\epsilon}.$$

1.3.3 (c)

Consider the two-norm regularization with the restricted hypothesis space:

$$\mathcal{H}_C = \{f(\boldsymbol{x}) = w_1 x_1 + w_2 x_2 | \boldsymbol{w} \in \mathbb{R}^2 \wedge w_1^2 + w_2^2 \leq C\}.$$

To find the minimum value of C, i first plug in the expression of w_1 and $w_2$from above into the regularization constraint:

$$w_1^2 + w_2^2 \leq C,$$
$$\Leftrightarrow \left(\hat{w}_1 - \hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)^2 + \left(\frac{\hat{w}_2}{\epsilon}\right)^2 \leq C,$$
$$\Leftrightarrow \hat{w}_1^2 + \left(\hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)^2 - 2\hat{w}_1 \hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{\hat{w}_2^2}{\epsilon^2} \leq C,$$
$$\Leftrightarrow \hat{w}_1^2 + \hat{w}_2^2 \frac{1-\epsilon^2}{\epsilon^2} - 2\hat{w}_1 \hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{\hat{w}_2^2}{\epsilon^2} \leq C,$$
$$\Leftrightarrow \hat{w}_1^2 + \hat{w}_2^2 \frac{2-\epsilon^2}{\epsilon^2} - 2\hat{w}_1 \hat{w}_2 \frac{\sqrt{1-\epsilon^2}}{\epsilon} \leq C.$$

The target function $f(\hat{\boldsymbol{x}})$ has $\hat{w}_1 = \hat{w}_2 = 1$. Substituting this in yields:

$$1 + \frac{2-\epsilon^2}{\epsilon^2} - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} \leq C,$$
$$\frac{2}{\epsilon^2} - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} \leq C,$$
$$\frac{2}{\epsilon^2}\left(1 - \epsilon\sqrt{1-\epsilon^2}\right) \leq C.$$

Thus we get the minimum value of C (the maximum amount of regularization):

$$C = \frac{2}{\epsilon^2}\left(1 - \epsilon\sqrt{1-\epsilon^2}\right).$$

1.3.4 (d)

when the correlation increases ($\epsilon \to 0$), the first part of the expression increases infinitely $\left(\frac{2}{\epsilon^2} \to \infty\right)$, and the second part converges ton 1 $\left(1 - \epsilon\sqrt{1-\epsilon^2} \to 1, \text{ as } \epsilon\sqrt{1-\epsilon^2} \to 0\right)$. This means that the minimum value of C increases ($C \to \infty$) as the correlation increases ($\epsilon \to 0$).

2 Hoeffding's Bound (15 points)

To calculate the probability that $S_n > 22$ with $n = 100$, i use the Hoffdings' inequality:

$$\mathbb{P}\left(S_n - E[S_n] \geq \varepsilon\right) \leq \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

X_i takes the values $X_i \in \{-2, 0.8, 1\}$. The lower bound is $a_i = -2$, and the upper bound is $b_i = 1$. The expected value $E[S_{100}]$ is given by:

$$E[S_{100}] = 100E[X_i] = 100 \cdot 1/3(-2 + 0.8 + 1) = -100\frac{0.2}{3} = -\frac{20}{3}.$$

This yields ϵ:

$$\varepsilon = 22 - E[S_{100}] = 22 - \left(-\frac{20}{3}\right) = \frac{86}{3}$$

I can now calculate the probability that $S_n > 22$ using the inequality:

$$\mathbb{P}(S_{100} > 22) \leq \mathbb{P}\left(S_{100} - E[S_{100}] \geq \frac{86}{3}\right) \leq \exp\left(-\frac{2 \cdot \left(\frac{86}{3}\right)^2}{100\left(1 - (-2)\right)^2}\right),$$

$$\Leftrightarrow \mathbb{P}(S > 22) \lesssim \exp(-1.8262) \approx 0.1610.$$

Thus, the probability that $S_n > 22$ is at most $\approx 16.1\%$.

3 Illustration of Markov's, Chebyshev's, and Hoeffding's Inequalities (25 points)

3.a

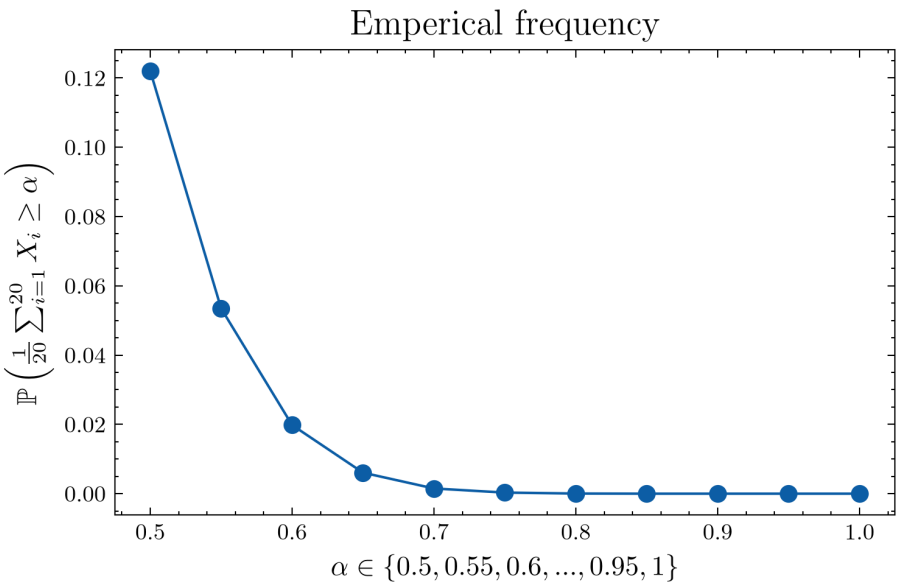
Python code used for generating data:

```
# Seed for random number generator
rng = np.random.default_rng(seed=999)

# Mean, number of random variables and number of experiments
mean = 0.35
n = 20
num_exp = 1_000_000
```

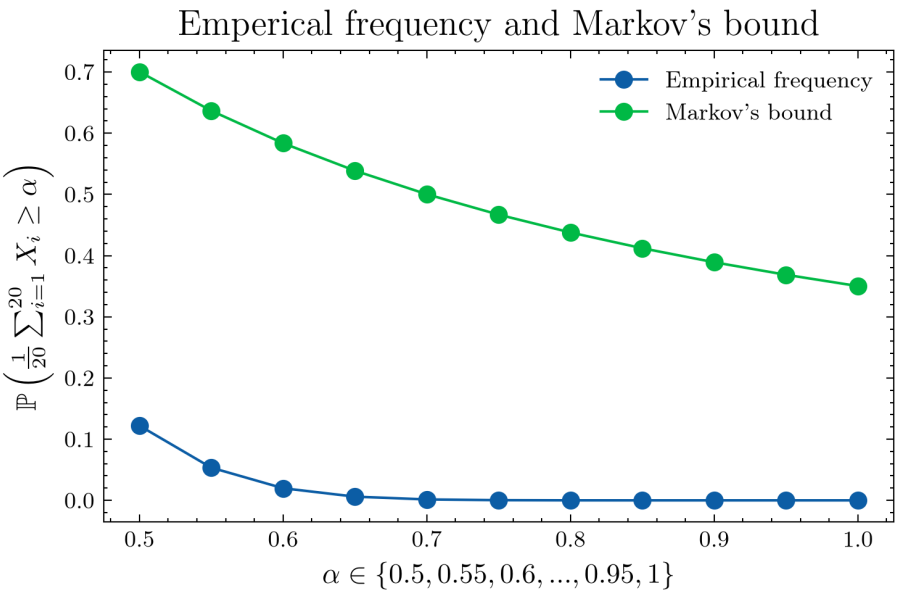
```
# Generate data (1.000.000 rows and 20 columns)
data_raw = rng.binomial(n=1, p=mean, size=(num_exp,n))
```

1.

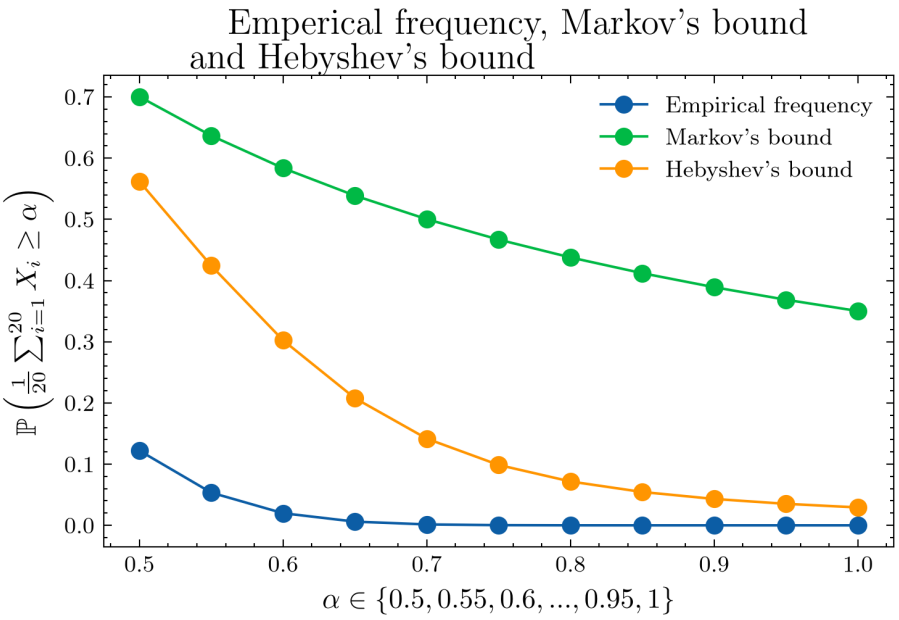


2. The chosen of α is sufficient, since X_i is binomial distributed. This means that the mean of 20 samples can only take values of $\frac{1}{20}\sum_{i=1}^{20} X_i \in [0, 0.05, 0.1, 0.15, 0.2, 025, \cdots, 0.95, 1]$.

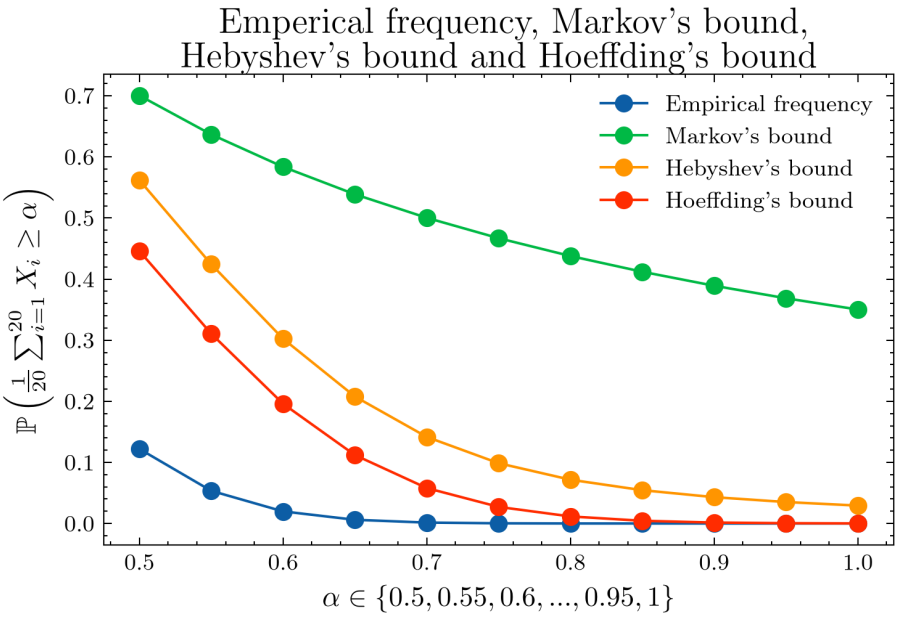
3.


Emperical frequency and Markov's bound
Empirical frequency
Markov's bound
$\mathbb{P}\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq \alpha\right)$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.5
0.6
0.7
0.8
0.9
1.0
$\alpha \in \{0.5, 0.55, 0.6, ..., 0.95, 1\}$

4.


Emperical frequency, Markov's bound
and Hebyshev's bound
Empirical frequency
Markov's bound
Hebyshev's bound
$\mathbb{P}\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq \alpha\right)$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.5
0.6
0.7
0.8
0.9
1.0
$\alpha \in \{0.5, 0.55, 0.6, ..., 0.95, 1\}$

5.



6. Comparing the four plots above, i can infer, that Markov's bound is worst at predicting the true probability $\mathbb{P}\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq \alpha\right)$, while Hebyshev's bound is closest to the true probability. The difference in the accuracy of predictions are a result of the progressively tighter assumptions required. Markov's bound only requires that $0 \leq E[X] < \infty$, Chebyshev's bound uses $Var(X)$, while Hoeffding's bound requires that $(X_1, X_2, \cdots, X_n)$ are iid., and that $X_i \in [a_i, b_i]$ are known.

7. To calculate the exact probability is the formula for Bernoulli variables:

$$P\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq \alpha\right) = \sum_{x=20\alpha}^{20} \binom{20}{x} (E[X_i])^x (1 - E[X_i])^{20-x},$$

where $\binom{20}{x} = \frac{20^x}{x!}$.

Using the formula gives the probability $P\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq 0.95\right)$:

$$P\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq 0.95\right) = \sum_{x=19}^{20} \binom{20}{x} (0.35)^x (1 - 0.35)^{20-x},$$

$$= 20(0.35)^{19}(0.65)^1 + (0.35)^{20} \approx 0.00000198\%,$$

and the probability $P\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq 1\right)$:

$$P\left(\frac{1}{20}\sum_{i=1}^{20} X_i \geq 0.20\right) = \sum_{x=20}^{20} \binom{20}{20} (0.35)^x (1 - 0.35)^{20-x},$$

$$= (0.35)^{20} \approx 0.0000000761\%.$$